A third magnetic Neumann level below the Dirichlet ground state on the disk

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Abstract

Let $H_b = (-i\nabla - b(-y,x)/2)^2$ be the spinless magnetic Laplacian on the unit disk, and let $\lambda_k(b)$ and $\mu_k(b)$ denote its Dirichlet and magnetic-Neumann eigenvalues, respectively, counted with multiplicity. We prove that $\mu_3(b) \le \lambda_1(b)$ for every $b > 0$. The previously unresolved part of this comparison is the bounded-field interval $0 < b \le 4$. There, the positive radial function $(1-r^2)e^{-r^2/2}$ gives a piecewise explicit lower bound on the Dirichlet ground energy. We compare that bound with the Rayleigh quotients of three mutually orthogonal magnetic-Neumann form-domain states. Two states work throughout the interval, while the third is chosen from one of the two angular-momentum states meeting the Dirichlet ground state at $b = 2$. The differences factor into elementary expressions whose signs are fixed on $0 < b \le 2$ and $2 \le b \le 4$. Min–max then avoids any spectral branch ordering or truncation. Combining this variational certificate with the published magnetic-Pauli counting argument for $b > 4$ proves the all-field statement.

Keywords: magnetic Laplacian; unit disk; Dirichlet–Neumann inequality; eigenvalue comparison; min–max principle

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1 Introduction

Inequalities between Dirichlet and Neumann spectra measure how much the low-energy state count changes when a hard boundary is replaced by a reflecting one. For the homogeneous magnetic Laplacian on a bounded convex planar domain, Lotoreichik proved the index-shifted comparison $\mu_{k+1} \le \lambda_k$ [3]. On the disk, numerical evidence in the same work suggested the sharper first-index relation $\mu_3(b) \le \lambda_1(b)$. The cited paper proved this relation for $b > 4$ and identified an exact three-branch crossing at $b = 2$, but left the continuous interval $0 < b \le 4$ at the numerical level. The nonmagnetic analogue and its geometric extensions form a related but distinct problem; see, for example, [4]. Recent detailed work on the magnetic Neumann disk has focused on its lowest eigenvalue and angular-momentum crossings [2].

The purpose of this note is to close the disk comparison by an exact low-field argument. Its ingredients are deliberately elementary: a ground-state representation for a positive Dirichlet comparison function, four explicit trial functions, angular orthogonality, and the min–max principle. In particular, the proof never has to determine which radial branch is the third eigenvalue.

Let

$$\mathbb{D} = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}, \qquad A_b(x,y) = \frac{b}{2}(-y,x), \qquad b > 0,$$

and set $H_b = (-i\nabla - A_b)^2$. The Dirichlet and magnetic-Neumann realizations are defined by the common quadratic form

$$q_b[u] = \int_{\mathbb{D}} |(-i\nabla - A_b)u|^2 \, dx\, dy$$

on $H_0^1(\mathbb{D})$ and $H^1(\mathbb{D})$, respectively. Their eigenvalues, repeated according to multiplicity, are denoted by $\lambda_k(b)$ and $\mu_k(b)$. Our main result is the following.

Theorem 1.1. *For every $b > 0$,*

$$\mu_3(b) \le \lambda_1(b).$$

The normalization is dimensionless. For a disk of radius R, dilation replaces b by bR^2 and multiplies both spectra by R^{-2}.

2 Angular decomposition and a Dirichlet lower bound

With the convention $u(r,\theta) = f(r)e^{im\theta}$, $m \in \mathbb{Z}$, the radial expression in $L^2((0,1); r\,dr)$ is

$$T_{b,m} = -\frac{d^2}{dr^2} - \frac{1}{r}\frac{d}{dr} + \left(\frac{m}{r} - \frac{br}{2}\right)^2. \tag{1}$$

The regular solution behaves as $O(r^{|m|})$ at the origin. Dirichlet data give $f(1) = 0$. Since A_b is tangent to the circle, magnetic-Neumann data reduce in this gauge to $f'(1) = 0$. Distinct angular sectors are orthogonal both for the L^2 inner product and for q_b.

We use the established disk fact that the Dirichlet ground energy is the ground energy of the $m = 0$ fiber [3, Sec. 5].

Lemma 2.1. *For $0 < b \le 4$,*

$$\lambda_1(b) \ge L(b) := \begin{cases} 5 + b^2/4, & 0 < b \le 2, \\ 6, & 2 \le b \le 4. \end{cases} \tag{2}$$

Proof. Put

$$\phi(r) = (1 - r^2)e^{-r^2/2}.$$

This function belongs to the radial Dirichlet form domain and is positive for $0 < r < 1$. Direct differentiation in (1) gives

$$\frac{T_{b,0}\phi}{\phi} = 6 + \frac{b^2 - 4}{4}r^2. \tag{3}$$

For a smooth radial form-core function f, integration by parts yields the ground-state representation

$$\begin{aligned} \int_0^1 \left(|f'|^2 + \frac{b^2 r^2}{4}|f|^2\right) r\,dr - \int_0^1 \frac{T_{b,0}\phi}{\phi}|f|^2 r\,dr \\ = \int_0^1 \phi^2 \left|\left(\frac{f}{\phi}\right)'\right|^2 r\,dr \ge 0. \end{aligned} \tag{4}$$

The inequality extends to the radial Dirichlet form domain by closure. For $0 < b \le 2$, the infimum of the right-hand side of (3) is $5 + b^2/4$; for $2 \le b \le 4$, it is 6. The radial characterization of the Dirichlet ground state now gives (2). □

3 An explicit three-dimensional trial space

For a nonzero $u \in H^1(\mathbb{D})$, write

$$\mathcal{R}_b[u] = \frac{q_b[u]}{\|u\|^2_{L^2(\mathbb{D})}}.$$

Two trial functions will be used for the whole interval $0 < b \le 4$:

$$u_0(r,\theta) = 1, \qquad u_1(r,\theta) = re^{i\theta}.$$

Direct radial integration gives

$$\mathcal{R}_b[u_0] = \frac{b^2}{8}, \qquad \mathcal{R}_b[u_1] = 4 - b + \frac{b^2}{6}. \tag{5}$$

Consequently $\mathcal{R}_b[u_0] \le 2$ and $\mathcal{R}_b[u_1] \le 4$ on this interval, so both are strictly below $L(b)$.

The third state changes at $b = 2$. Define

$$v_-(r,\theta) = re^{-r^2/2}e^{-i\theta},$$
$$v_+(r,\theta) = r^2(3 - r^2)e^{-r^2/2}e^{2i\theta}.$$

Both are regular $H^1(\mathbb{D})$ functions. At $b = 2$, their radial parts satisfy $T_{2,-1}v_- = 6v_-$ and $T_{2,2}v_+ = 6v_+$, respectively, as well as the Neumann condition at $r = 1$ [3, Sec. 5]. Set

$$\rho_\pm = \frac{\int_0^1 r^2|v_\pm(r)|^2 r\,dr}{\int_0^1 |v_\pm(r)|^2 r\,dr}. \tag{6}$$

Only the exact support bound

$$0 < \rho_\pm < 1 \tag{7}$$

will be needed.

Lemma 3.1. *The trial state v_- satisfies $\mathcal{R}_b[v_-] \le L(b)$ for $0 < b \le 2$, and v_+ satisfies $\mathcal{R}_b[v_+] \le L(b)$ for $2 \le b \le 4$.*

Proof. Changing b in (1) changes only the potential

$$\left(\frac{m}{r} - \frac{br}{2}\right)^2 = \frac{m^2}{r^2} - mb + \frac{b^2r^2}{4}.$$

Using the exact eigenvalue 6 at $b = 2$ therefore gives

$$\mathcal{R}_b[v_-] = 6 + (b - 2) + \frac{b^2 - 4}{4}\rho_-. \tag{8}$$

For $0 < b \le 2$, subtraction of the first branch of (2) factorizes as

$$\mathcal{R}_b[v_-] - \left(5 + \frac{b^2}{4}\right) = \frac{b - 2}{4}\left((\rho_- - 1)b + 2(1 + \rho_-)\right). \tag{9}$$

The expression in parentheses is positive: it decreases with b, and at $b = 2$ it equals $4\rho_- > 0$. Thus (9) is nonpositive.

For the $m = 2$ state, the same potential shift gives

$$\mathcal{R}_b[v_+] - 6 = (b - 2)\left(-2 + \frac{b + 2}{4}\rho_+\right). \tag{10}$$

When $2 \le b \le 4$, (7) implies

$$-2 + \frac{b+2}{4}\rho_+ < -2 + \frac{b+2}{4} \le -\frac{1}{2}.$$

Hence (10) is also nonpositive. □

Proof of Theorem 1.1 for $0 < b \le 4$. For $0 < b \le 2$, take

$$\mathcal{V}_b = \operatorname{span}\{u_0, u_1, v_-\},$$

whose elements have angular momenta $0, 1, -1$. For $2 \le b \le 4$, take instead

$$\mathcal{V}_b = \operatorname{span}\{u_0, u_1, v_+\},$$

with angular momenta $0, 1, 2$. In either case $\dim \mathcal{V}_b = 3$ and angular orthogonality diagonalizes both the norm and the quadratic form. The Rayleigh quotient of every nonzero element of $\mathcal{V}_b$ is thus a weighted average of the three individual quotients. Equations (5), (2), and Lemma 3.1 show that its maximum is at most $\lambda_1(b)$. The magnetic-Neumann min–max principle then yields

$$\mu_3(b) \le \max_{0 \ne u \in \mathcal{V}_b} \mathcal{R}_b[u] \le \lambda_1(b).$$

The switch point $b = 2$ and the endpoint $b = 4$ are included. □

4 The high-field interval

For completeness, we recall how the published high-field result joins the preceding argument. The normalized flux of the constant field through the unit disk is

$$\Phi = \frac{1}{2\pi} \int_{\mathbb{D}} b \, dx \, dy = \frac{b}{2}.$$

The magnetic-Pauli counting theorem of Fournais, Frank, Goffeng, Kachmar, and Sundqvist [1, Prop. 2.1], specialized to the disk in [3, Sec. 5], implies that $b > 4$ gives at least three magnetic-Neumann eigenvalues strictly below b. Thus

$$\mu_3(b) < b \qquad (b > 4). \tag{11}$$

On the Dirichlet form domain, if $\Pi_x = -i\partial_x + by/2$ and $\Pi_y = -i\partial_y - bx/2$, then $[\Pi_x, \Pi_y] = ib$ and integration by parts gives

$$q_b[u] - b\|u\|^2 = \|(\Pi_x + i\Pi_y)u\|^2 \ge 0.$$

Consequently $\lambda_1(b) \ge b$. Together with (11), this proves $\mu_3(b) < \lambda_1(b)$ for $b > 4$. Combining it with the low-field proof completes Theorem 1.1.

5 Interpretation and limitations

The theorem is a spectral statement for two self-adjoint boundary realizations of the same spinless constant-field Hamiltonian. It says that replacing the hard-wall boundary by magnetic reflection puts at least three levels, counted with multiplicity, at or below the hard-wall ground energy. The proof is nonperturbative and uses no finite angular-momentum cutoff, radial discretization, or numerical ordering of crossing branches.

The circular geometry is essential to the present certificate: it supplies the angular-sector orthogonality and the explicit states at $b = 2$. No conclusion is asserted here for noncircular domains, variable fields, Robin conditions, Pauli operators with a Zeeman term, or comparisons involving higher indices.

Statements and declarations

Data and code availability. The result is analytic and uses no empirical data. The accompanying source package contains a symbolic verification script and its recorded output for the differentiations, Rayleigh quotients, and factorizations used above.

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